
Lab 2 – Cap Database

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1 Select * Queries

```
1 SELECT *
2 FROM PEOPLE
```

pid	prefix	firstname	lastname	suffix	homecity	dob
1	Dr. (Hon)	Maynard	Ferguson	Esq.	Montreal	1928-05-04
2	Ms.	Bria	Skonberg		Chilliwack	1987-12-29
3	Mr.	Miles	Davis		Alton	1926-05-26
4	Mr.	Doc	Severinsen		Arlington	1927-07-07
5	Mr.	Louis	Armstrong	Esq.	New Orleans	1901-08-04
6	Ms.	Tine	Helseth		Oslo	1987-08-18
7	Dr.	Cynthia	Robinson	MD	Sacramento	1944-01-12
8	Dr.	James	Morrison	Ph.D.	Oslo	1962-11-11
10	Mr.	Dizzy	Gillespie	III	Montreal	1917-10-21

```
1 SELECT *
2 FROM Customers
```

pid	paymentterms	discountpct
1	Net 30	21.12
4	Net 15	2.47
5	In Advance	5.05
7	On Receipt	2.00
8	Net 30	10.01

```

1 SELECT *
2 FROM Agents

```

pid	paymentterms	commissionpct
2	Quarterly	5.00
3	Annually	10.00
5	Monthly	1.00
6	Weekly	2.00

```

1 SELECT *
2 FROM Products

```

prodid	name	city	qtyonhand	priceusd
p01	Heisenberg Compensator	Dallas	47	67.76
p02	Universal Translator	Newark	2399	51.50
p03	Apple //+	Duluth	1979	65.02
p04	LCARS module	Duluth	3	17.01
p05	Denis Wick Valve Oil	Dallas	8675309	16.61
p06	PDP-11 operator panel	Beijing	88	88.00
p07	Flux Capacitor	Newark	1007	1.00
p08	HAL 9000 memory chip	Newark	200	1.25
p09	Bach Stradivarius 37	Montreal	1	37900.42

```

1 SELECT *
2 FROM Orders

```

ordernum	dateordered	custid	agentid	prodid	quantityordered	totalusd
1011	2024-01-22	1	2	p01	1100	58794.00
1012	2023-01-23	4	3	p03	1200	76096.81
1015	2022-01-23	5	3	p05	1000	15771.20
1016	2021-01-23	8	3	p01	1000	60977.22
1017	2023-02-14	1	3	p03	500	25643.98
1018	2023-02-14	1	3	p04	600	8050.49
1019	2023-02-14	1	2	p02	400	16249.28
1020	2023-02-14	4	5	p07	600	585.18
1021	2023-02-14	4	5	p01	1000	66086.33
1022	2023-03-15	1	3	p06	450	31236.48
1023	2023-03-15	1	2	p05	500	6550.98
1024	2023-03-15	5	2	p01	880	56671.55
1025	2022-04-01	8	3	p07	888	799.11
1026	2022-05-04	8	5	p03	808	47277.29

2 Primary key, Candidate key, and Superkey

A super key is a set of fields (at least one) of a table which uniquely identifies each row. A candidate key is the minimal length of super keys which uniquely identifies each row. Therefore, every candidate key is a super key but not every super key is a candidate key. Any key which is of multiple fields is known as a composite key. A primary key is the candidate key which is chosen to

uniquely identify the row. It is also very important to consider the business rules associated with the database when selecting keys. A primary key should be functionally unique within its domain.

3 Data Types in Tables

Without data types whatever memory stored is useless series of binary digits. Giving your datatypes the most strict type that possible is crucial to getting the most information out of your data. Imagine you have to store students as a table in your database let's break down the data types that would need to be stored. Columns and attributes such as first name and last name are fairly obviously strings of some length. However, it is objectively better to separate these in most use cases into the two fields instead of something like "name" as being specific extracts more information. Another field you would likely want to include grade year. If these are students in a highschool, for example, it may be tempting to store their grade as a string. This ends up being a terrible idea: when data gets entered it has full access to all string representation. Should people enter a freshmen as freshmen, Freshmen, freshman, Freshman, freshy, first-year, etc. It would be much better to store it has a enumeration of possibilities. In college, this representation would be less useful as grade-level does not carry as much information as it does for high school. It may be better to not even have grade-level stored and instead have it be derived from credits earned or courses passed. There is much more information you could include such as gender, race, address, previous schooling etc. Each of these bits of information needs to be meticulously mapped to accurate data types when translating them to a column or columns.

4 Relational Rules

Making databases ensure relational rules upholds the referential integrity of the data helping to render information. It is good practice to follow these rules as well as infusing business rules into the database itself when possible.

4.1 First Normal Form

The first normal form maintains that there are no multi-valued structures within a field or fields which then relate to structure within the business which semantically should be separated in structure. If a field has some internal structure that can be practically separated it should be.

4.2 Access Rows by Content Only

As the database's internal structure is hidden from the user or programmers, accessing data using concepts such as index does not make sense. This makes

sense because in postgres, for example, all tables and columns are themselves stored in a database. This means there is no concept of index as the underlying structure likely used is a set.

4.3 Unique Rows

Every row within a table should have a unique primary key rules of which have been previously mentioned. This is need to differinate and assert no duplication of similar or the same thing within the database.